

Homework Problems

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1 Overview

The following document is a compilation of various calculations, computations, and exercises I completed as part of orals prep. The work here is not meant to be an extensive account of my work, by any means.

2 Functional and Harmonic Analysis

2.1 Stein & Shakarchi

Problem 1.1. Consider $L^p = L^p(\mathbb{R}^d)$ with Lebesgue measure. Let $f_0(x) = |x|^{-\alpha}$ if $|x| < 1$, $f_0(x) = 0$ for $|x| \geq 1$; also let $f_\infty(x) = |x|^{-\alpha}$ if $|x| \geq 1$, $f_\infty(x) = 0$ when $|x| < 1$. Show that:

- (a) $f_0 \in L^p$ if and only if $p\alpha < d$.
- (b) $f_\infty \in L^p$ if and only if $d < p\alpha$.
- (c) What happens if in the definitions of f_0 and f_∞ we replace $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2/|x|))$ for $|x| < 1$, and $|x|^{-\alpha}$ by $|x|^{-\alpha}/(\log(2|x|))$ for $|x| \geq 1$?

- (a) Let $p \geq 1$, and define $f_0(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ to be the measurable function

$$f_0(x) = \begin{cases} |x|^{-\alpha}, & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

Then $f_0(x) \in L^p$ if and only if $\int_{|x| \leq 1} |f_0(x)|^p dx = \int_{|x| \leq 1} |x|^{-p\alpha} dx < \infty$. To evaluate this integral, we will use integration in polar coordinates. Using Theorem 2.49 from Folland, one has

$$\int_{|x| \leq 1} \frac{1}{|x|^{p\alpha}} dx = \Gamma(S^{d-1}) \cdot \int_0^1 r^{-p\alpha+d-1} dr.$$

By calculus in one dimension, we know that the integral on the right side is finite if and only if $-p\alpha + d - 1 > -1$, which implies $p\alpha < d$. Thus, $f_0 \in L^p$ if and only if $p\alpha < d$.

- (b) Define $f_\infty(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ to be measurable function

$$f_\infty(x) = \begin{cases} 0, & |x| < 1 \\ |x|^{-\alpha}, & |x| \geq 1. \end{cases}$$

Then $f_\infty(x) \in L^p$ if and only if $\int_{|x| \geq 1} |f_\infty(x)|^p dx = \int_{|x| \geq 1} |x|^{-p\alpha} dx < \infty$. Using Theorem 2.49 from Folland, one has

$$\int_{|x| \geq 1} \frac{1}{|x|^{p\alpha}} dx = \Gamma(S^{d-1}) \cdot \int_1^\infty r^{-p\alpha+d-1} dr.$$

By calculus in one dimension, we know that the integral on the right side is finite if and only if $-p\alpha + d - 1 < -1$, which implies $p\alpha > d$. Thus, $f_\infty \in L^p$ if and only if $p\alpha > d$.

- (c) Now define $f_0(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ to be the modified function

$$f_0(x) = \begin{cases} \frac{|x|^{-\alpha}}{\log(2/|x|)}, & |x| < 1 \\ 0, & |x| \geq 1. \end{cases}$$

As in (a), $f_0(x) \in L^p$ if and only if

$$\int_0^1 \frac{r^{-p\alpha+d-1}}{\log(2/r)^p} dr < \infty.$$

Let $k := -p\alpha + d - 1$, and define $u = \log(2/r)$. Then the above integral transforms into

$$\int_0^1 \frac{r^{-p\alpha+d-1}}{\log(2/r)^p} dr = - \int_{\log(2)}^\infty \frac{2^k e^{-ku}}{u^p} (-2e^{-u} du) = 2^{k+1} \int_{\log(2)}^\infty \frac{e^{-(k+1)u}}{u^p} du.$$

In the limit $u \rightarrow \infty$, if $p > 1$, the integrand decays to zero provided $k + 1 \geq 0$. Thus, $f_0(x) \in L^p$ if and only if $k \geq -1$, which is equivalent to requiring $p\alpha \leq d$. If $p = 1$, then $f_0(x) \in L^p$ if and only if $p\alpha < d$. Now define the modified function $f_\infty(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ by

$$f_\infty(x) = \begin{cases} 0, & |x| < 1 \\ \frac{|x|^{-\alpha}}{\log(2|x|)}, & |x| \geq 1. \end{cases}$$

As in (b), $f_\infty(x) \in L^p$ if and only if the integral

$$\int_1^\infty \frac{r^{-p\alpha+d-1}}{\log(2r)} dr < \infty.$$

Taking $u = \log(2r)$ this time, the above integral transforms into

$$\int_1^\infty \frac{r^{-p\alpha+d-1}}{\log(2r)^p} dr = \int_{\log(2)}^\infty \frac{2^{-k} e^{ku}}{u^p} (2^{-1} e^u du) = 2^{-(k+1)} \int_{\log(2)}^\infty \frac{e^{(k+1)u}}{u^p} du.$$

Here, we must consider two cases. First, suppose $p > 1$. Then the integrand decays to zero provided $k+1 \leq 0$ (corresponding to $p\alpha \geq d$). Thus $f_\infty(x) \in L^p$ for $p > 1$ if and only if $p\alpha \geq d$. On the other hand, if $p = 1$, then the integral is finite if and only if $k+1 \not\leq 0$, which corresponds to $p\alpha < d$. Thus,

$$f_\infty(x) \in L^p(\mathbb{R}^d) \text{ if and only if } \begin{cases} p\alpha \geq d, & p > 1 \\ \alpha > d, & p = 1. \end{cases}$$

Problem 1.8. Suppose $1 \leq p < \infty$, and that $\mathbb{R}^d$ is equipped with Lebesgue measure. Show that if $f \in L^p(\mathbb{R}^d)$, then

$$\|f(x+h) - f(x)\|_{L^p} \rightarrow 0 \quad \text{as } |h| \rightarrow 0.$$

Prove that this fails when $p = \infty$.

[Hint: By the previous exercise, the continuous functions with compact support are dense in $L^p(\mathbb{R}^d)$ for $1 \leq p < \infty$.]

Fix $1 \leq p < \infty$, and assume that $\mathbb{R}^d$ is equipped with Lebesgue measure. Further fix some $f \in L^p(\mathbb{R}^d)$. By the density of $C_c(\mathbb{R}^d)$ in $L^p(\mathbb{R}^d)$, there exists a sequence $\{g_j\}_{j=1}^\infty$ of compactly supported continuous functions so that $g_j \rightarrow f$ under the L^p -norm. First, we will show that for every j , $\|g_j(h+\cdot) - g_j\|_p \rightarrow 0$ as $h \rightarrow 0$. Without loss of generality, let $|h| \leq R$ for some $R > 0$. Then by continuity of g_j , $|g_j(x+h) - g_j(x)|^p \rightarrow 0$ pointwise a.e. as $|h| \rightarrow 0$. On the other hand, by continuity of g_j and the extreme value theorem, one has

$$|g_j(x+h) - g_j(x)|^p \leq 2^{p-1}(|g_j(x+h)|^p + |g_j(x)|^p) \leq M \chi_{K+B_R(0)} \in L^1(\mathbb{R}^d),$$

where $M = 2^p \cdot (\max_K |g_j(x)|)^p$. Thus, by the Dominated Convergence Theorem,

$$\int_{\mathbb{R}^d} |g_j(x+h) - g_j(x)|^p dx \xrightarrow{|h| \rightarrow 0} 0.$$

Now fix $\varepsilon > 0$, j large enough so that $\|f - g_j\|_p^p < \varepsilon/3$, and let $\delta > 0$ so that for all $|h| < \delta$, one has $\|g_j(h+\cdot) - g_j\|_p^p < \varepsilon/3$. Then we observe that

$$\begin{aligned} \|f(h+\cdot) - f\|_p^p &= \int_{\mathbb{R}^d} |f(x+h) - f(x)|^p dx \\ &\leq C \cdot \left[\int_{\mathbb{R}^d} |f(x+h) - g_j(x+h)|^p dx + \int_{\mathbb{R}^d} |g_j(x+h) - g_j(x)|^p dx + \int_{\mathbb{R}^d} |g_j(x) - f(x)|^p dx \right] \\ &= 2C \|f - g_j\|_p^p + C \cdot \|g_j(h+\cdot) - g_j\|_p^p \\ &< C\varepsilon. \end{aligned}$$

where in going from the second to the third line, we used translation invariance of the L^p norms. Since the left side is independent of ε , taking the limit $\varepsilon \rightarrow 0$ shows that $\|f(h+\cdot) - f\|_p^p \rightarrow 0$ as $|h| \rightarrow 0$.

Now consider the case of $p = \infty$. Let $f \in L^\infty(\mathbb{R}^d)$ be the following function:

$$f(x) = \begin{cases} 1, & |x| \geq 1 \\ -1, & |x| < 1. \end{cases}$$

Then for any $|h| > 0$, one has that $\|f(h+\cdot) - f\|_{L^\infty} = 2$ so that this norm cannot converge to zero.

Problem 1.12. Suppose the measure space (X, μ) is separable as defined in Exercise 10. Let $1 \leq p < \infty$ and $1/p + 1/q = 1$. A sequence $\{f_n\}$ with $f_n \in L^p$ is said to converge to $f \in L^p$ **weakly** if

$$\int f_n g \, d\mu \rightarrow \int f g \, d\mu \quad \text{for every } g \in L^q.$$

- (a) Verify that if $\|f - f_n\|_{L^p} \rightarrow 0$, then f_n converges to f weakly.
- (b) Suppose $\sup_n \|f_n\|_{L^p} < \infty$. Then, to verify weak convergence, it suffices to check the above convergence for a dense subset of functions g in L^q .
- (c) Suppose $1 < p < \infty$. Show that if $\sup_n \|f_n\|_{L^p} < \infty$, then there exists $f \in L^p$ and a subsequence $\{n_k\}$ so that f_{n_k} converges weakly to f .

- (a) Assume that $\|f - f_n\|_{L^p} \rightarrow 0$. Fix an arbitrary $g \in L^q$. Then by Hölder's inequality,

$$\left| \int f_n g \, d\mu - \int f g \, d\mu \right| \leq \int |f_n - f| |g| \, d\mu \leq \|f_n - f\|_p \|g\|_q \xrightarrow{n \rightarrow \infty} 0.$$

Since $g \in L^q$ was arbitrary, we conclude that f_n converges to f weakly.

- (b) Suppose $\sup_n \|f_n\|_{L^p} < \infty$, and further assume that the above result holds for some dense subset $\mathcal{A}$ in L^q . Fix an arbitrary $g \in L^q$, and let $\{g_j\}_{j=1}^\infty \subset \mathcal{A}$ so that $\|g - g_j\|_q \rightarrow 0$ as $j \rightarrow \infty$. Then

$$\begin{aligned} \left| \int f_n g - \int f g \right| &= \left| \int f_n g - \int f_n g_j + \int f_n g_j - \int f g_j + \int f g_j - \int f g \right| \\ &\leq \int |f_n| |g - g_j| + \left| \int f_n g_j - \int f g_j \right| + \int |f| |g - g_j| \\ &\leq \|f_n\|_{L^p} \|g - g_j\|_{L^q} + \left| \int f_n g_j - \int f g_j \right| + \|f\|_{L^p} \|g - g_j\|_{L^q} \\ &\leq \sup_n \|f_n\|_{L^p} \|g - g_j\|_{L^q} + \left| \int f_n g_j - \int f g_j \right| + \|f\|_{L^p} \|g - g_j\|_{L^q}. \end{aligned}$$

Thus, taking the limit $n \rightarrow \infty$ on both sides, and using the fact that for each j , the second term vanishes under the limit by hypothesis, one has

$$\lim_{n \rightarrow \infty} \left| \int f_n g - \int f g \right| \leq \sup_n \|f_n\|_{L^p} \|g - g_j\|_{L^q} + \|f\|_{L^p} \|g - g_j\|_{L^q}.$$

Taking the limit $j \rightarrow \infty$ shows that $\int f_n g \rightarrow \int f g$ so that f_n converges to f weakly.

- (c) Suppose $1 < p < \infty$, and assume $\sup_n \|f_n\|_{L^p} < \infty$. First, since X was assumed to be separable, then by Exercise 10(b), one has that $L^q(X, \mu)$ is separable. So let $\{g_j\}_{j=1}^\infty \subset L^q(X, \mu)$ be a countable dense collection. We consider the sequence $\{\int f_n g_1 \, d\mu\}_{n=1}^\infty \subset \mathbb{R}$. First, it follows that this sequence is uniformly bounded: for every n ,

$$\left| \int f_n g_1 \, d\mu \right| \leq \int |f_n| |g_1| \, d\mu \leq \|f_n\|_{L^p} \|g_1\|_{L^q} \leq \sup_n \|f_n\|_{L^p} \|g_1\|_{L^q} < \infty.$$

Thus there exists a subsequence $\{f_{n_k}^{(1)}\}_{k=1}^\infty$ so that $\{\int f_{n_k}^{(1)} g_1 \, d\mu\}_{k=1}^\infty$ converges. Now consider the sequence $\{\int f_{n_k}^{(1)} g_2 \, d\mu\}_{k=1}^\infty$. As before, this sequence is uniformly bounded so that there exists a further subsequence $\{f_{n_k}^{(2)}\}_{k=1}^\infty$ such that $\{\int f_{n_k}^{(2)} g_2 \, d\mu\}_{k=1}^\infty$ converges. By construction, $\{\int f_{n_k}^{(2)} g_1 \, d\mu\}_{k=1}^\infty$ also converges. In this manner, for each j , we can obtain a subsequence $\{f_{n_k}^{(j)}\}_{k=1}^\infty$ so that $\{\int f_{n_k}^{(j)} g_\ell \, d\mu\}_{k=1}^\infty$ converges for all $\ell \leq j$. Now construct the sequence $\{f_{n_k}\}_{k=1}^\infty = \{f_{n_k}^{(j)}\}_{j=1}^\infty$. By construction, $\{\int f_{n_k} g_j \, d\mu\}_{k=1}^\infty$ converges for all j . Now we claim that $\{\int f_{n_k} g \, d\mu\}_{k=1}^\infty$ converges for all $g \in L^q$. Let $\varepsilon > 0$ be fixed, $M := \sup_n \|f_n\|_{L^p} < \infty$, j large enough so that $\|g - g_j\|_{L^q} < \varepsilon/(3M)$, and N large enough so that for all $k, m \geq N$, $|\int f_{n_k} g_j - \int f_{n_m} g_j| < \varepsilon/3$. Then for all $k, m \geq N$, one has

$$\begin{aligned} \left| \int f_{n_k} g - \int f_{n_m} g \right| &\leq 2 \sup_n \|f_n\|_{L^p} \cdot \|g - g_j\|_{L^q} + \left| \int f_{n_k} g_j - \int f_{n_m} g_j \right| \\ &< \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Thus taking $\varepsilon \rightarrow 0$ concludes the proof of the above claim. Now define the linear functional $T : L^q \rightarrow \mathbb{R}$ as follows: for any $g \in L^q$, let

$$T(g) := \lim_{n_j \rightarrow \infty} \int f_{n_j} g.$$

It is straightforward to see that this functional is bounded. Thus since $1 < p, q < \infty$ and p, q are conjugates, by the Riesz Representation Theorem, there exists a unique $f \in L^p$ so that

$$\int f g = T(g) = \lim_{n_j \rightarrow \infty} \int f_{n_j} g \quad (g \in L^q(X, \mu)).$$

Thus the proof concludes.

Problem 2.2. The following are simple generalizations of the Hausdorff-Young inequalities.

- (a) Suppose $\{\varphi_n\}$ is an orthonormal sequence on $L^2(X, \mu)$. Assume also that $|\varphi_n(x)| \leq M$ for all n . If $a_n = \int f \bar{\varphi}_n d\mu$, then $\|a_n\|_{L^q} \leq M^{(2/p)-1} \|f\|_{L^p(X)}$, $1 \leq p \leq 2$, $1/p + 1/q = 1$.
- (b) Suppose $f \in L^p$ on the torus $\mathbb{T}^d$, and $a_n = \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x} dx$, $n \in \mathbb{Z}^d$. Then $\|\{a_n\}\|_{L^q(\mathbb{Z}^d)} \leq \|f\|_{L^p(\mathbb{T}^d)}$, where $1/q \leq 1 - 1/p$.

- (a) We initially define the operator T on simple functions f by $Tf = \{a_n\}$, where

$$a_n := \int f \bar{\varphi}_n d\mu.$$

Then clearly, for $(p_0, q_0) = (1, \infty)$, one has that since for each n ,

$$|a_n| \leq \int |f| |\bar{\varphi}_n| d\mu \leq \int M |f| = M \|f\|_{L^1(X)},$$

$\|a_n\|_{L^\infty} = \sup |a_n| \leq M \|f\|_{L^1(X)}$. Thus by Proposition 5.4, we conclude that there exists a unique extension T to all of $L^1(X)$ satisfying the same bound. Likewise, for $(p_1, q_1) = (2, 2)$ and any simple function f , Bessel's inequality implies that

$$\|a_n\|_{L^2} = \left(\sum_{n=1}^{\infty} |a_n|^2 \right)^{1/2} \leq \|f\|_{L^2(X)} = M^{\frac{2}{2}-1} \|f\|_{L^2(X)}.$$

Thus by Proposition 5.4, there exists a unique extension T to all of $L^2(X)$ satisfying the same bound. Then, the Riesz-Thorin Interpolation Theorem tells us that $\|a_n\|_{L^q} \leq M \|f\|_{L^p(X)}$ where

$$\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1} = 1 - \frac{1}{2}t, \quad \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1} = \frac{1}{2}t,$$

where $t \in [0, 1]$ (thus corresponding to $1 \leq p \leq 2$, $p^{-1} + q^{-1} = 1$), and where

$$M \leq M^{1-t} = M^{2(1-(1/2)t)-1} = M^{\frac{2}{p}-1}.$$

- (b) As before, initially define the operator T on simple functions f on $\mathbb{T}^d$ by $Tf = \{a_n\}$, where for each $n \in \mathbb{Z}^d$,

$$a_n := \int_{\mathbb{T}^d} f(x) e^{-2\pi i n \cdot x} dx.$$

Clearly for $(p_0, q_0) = (1, \infty)$, one has

$$|a_n| \leq \int_{\mathbb{T}^d} |f(x)| = \|f\|_{L^1(\mathbb{T}^d)},$$

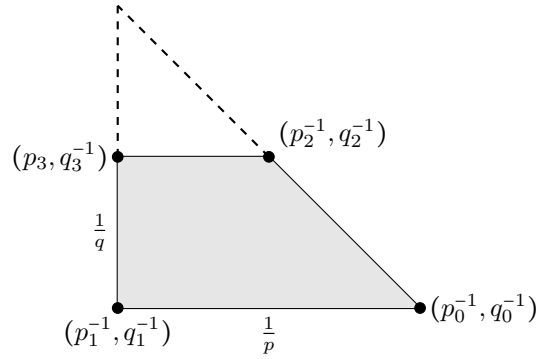
so that

$$\|\{a_n\}\|_{L^\infty(\mathbb{Z}^d)} = \sup_n |a_n| \leq \|f\|_{L^1(\mathbb{T}^d)}.$$

Thus Proposition 5.4 gives a unique extension of T to all $L^1(\mathbb{T}^d)$ that satisfies the same bound. Now consider $(p_1, q_1) = (\infty, \infty)$. Then for any $n \in \mathbb{Z}^d$, one has that

$$|a_n| \leq \int_{\mathbb{T}^d} |f(x)| \leq \|f\|_{L^\infty(\mathbb{T}^d)},$$

where we assume that dx denotes the normalized measure on $\mathbb{T}^d$. Thus taking the supremum over all $n \in \mathbb{Z}^d$, and using Proposition 5.4 concludes the claim. Likewise, one can show this property holds for the pair $(p_2, q_2) = (2, 2)$. Since $\mathbb{T}^d$ has finite measure, it follows that for all $p \geq 2$ (i.e., $1/p \leq 1/2$), $L^p(\mathbb{T}^d) \hookrightarrow L^2(\mathbb{T}^d)$. Thus, using Proposition 5.4, we also get the endpoint $(p_3, q_3) = (p, 2)$ for all $p \geq 2$. The Riesz diagram as of now is given by



Problem 2.3. Check that an inequality of the form $\|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A \|f\|_{L^p(\mathbb{R}^d)}$ (holding for all simple functions f) is possible if and only if $1/p + 1/q = 1$.
 [Hint: Let $f_r(x) = f(rx)$, $r > 0$. Then $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$.]

Let f be an arbitrary fixed simple function. As suggested by the hint, let $r > 0$, and define $f_r(x) = f(rx)$ so that $\hat{f}_r(\xi) = \hat{f}(\xi/r)r^{-d}$. We note that f_r is also a simple function. We now compute,

$$\begin{aligned} \|\hat{f}_r\|_{L^q(\mathbb{R}^d)} &= \left(\int_{\mathbb{R}^d} |\hat{f}(\xi/r)|^q r^{-dq} d\xi \right)^{1/q} \\ &= r^{-d} \left(\int_{\mathbb{R}^d} |\hat{f}(\xi/r)|^q d\xi \right)^{1/q} \\ &= r^{-d} \left(\int_{\mathbb{R}^d} |\hat{f}(\eta)|^q r^d d\eta \right)^{1/q} \\ &= r^{d(-1+1/q)} \|\hat{f}\|_{L^q(\mathbb{R}^d)}. \end{aligned}$$

Likewise,

$$\begin{aligned} \|f_r\|_{L^p(\mathbb{R}^d)} &= \left(\int_{\mathbb{R}^d} |f(rx)|^p dx \right)^{1/p} \\ &= \left(\int_{\mathbb{R}^d} |f(y)| r^{-d} dy \right)^{1/p} = r^{-d/p} \|f\|_{L^p(\mathbb{R}^d)}. \end{aligned}$$

So we have the following: the above estimate holds for all simple functions f if and only if for any $r > 0$,

$$\|\hat{f}_r\|_{L^q(\mathbb{R}^d)} \leq A \|f\|_{L^p(\mathbb{R}^d)} \quad \text{and} \quad \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq A \|f\|_{L^p(\mathbb{R}^d)}.$$

But our work above shows that the first inequality is equivalent to the estimate

$$r^{-d+d/q} \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq r^{-d/p} \|f\|_{L^p(\mathbb{R}^d)} \iff \|\hat{f}\|_{L^q(\mathbb{R}^d)} \leq r^{d(1-1/q-1/p)} A \|f\|_{L^p(\mathbb{R}^d)}.$$

Thus since $r > 0$ is arbitrary, the above estimate is consistent with the two inequality we have above if and only if $1 - 1/q - 1/p = 0$ iff $1/q + 1/p = 1$.

Problem 2.10. Suppose $f \in L^p(\mathbb{R})$. Verify that:

- (a) $\|f * \mathcal{P}_y\|_{L^p(\mathbb{R})} \leq \|f\|_{L^p}$, $1 \leq p \leq \infty$.
- (b) $f * \mathcal{P}_y \rightarrow f$, as $y \rightarrow 0$ in the L^p norm when $1 \leq p < \infty$.

(a) Let $\mathcal{P}_y(x)$ denote the **Poisson kernel**,

$$\mathcal{P}_y(x) := \frac{1}{\pi} \frac{y}{x^2 + y^2}.$$

Let $p_0 = 1$, and define the operator T on $L^{p_0}(\mathbb{R})$, initially just on the set of all simple functions, by $Tf := (f * \mathcal{P}_y)(x)$. First, if $y = 0$, then $\mathcal{P}_y(x) \equiv 0$ so that $Tf \equiv 0$, and the result holds trivially. So assume $y \neq 0$ (wlog, assume $y > 0$). Then we compute,

$$\begin{aligned} \|\mathcal{P}_y\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} \frac{1}{\pi} \frac{|y|}{x^2 + y^2} dx \\ &= \frac{1}{\pi} \frac{|y|}{y} \arctan\left(\frac{x}{y}\right) \Big|_{-\infty}^{\infty} = \frac{1}{\pi} \frac{|y|}{y} \cdot \left[\frac{\pi}{2} + \frac{\pi}{2}\right] = \frac{|y|}{y} = 1. \end{aligned}$$

Thus for $p_0 = 1$, we have

$$\begin{aligned} \|Tf\|_{L^1(\mathbb{R})} &= \int_{\mathbb{R}} \left| \frac{1}{\pi} \int_{\mathbb{R}} \frac{f(x-t)y}{y^2 + t^2} dt \right| dx \\ &\leq \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x-t)| \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dx dt \\ &= \int_{\mathbb{R}} \|f(\cdot - t)\|_{L^1(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &= \int_{\mathbb{R}} \|f\|_{L^1(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt = \|f\|_{L^1(\mathbb{R})} \|\mathcal{P}_y\|_{L^1(\mathbb{R})} = \|f\|_{L^1(\mathbb{R})}, \end{aligned}$$

where the second line follows from Minkowski's inequality, and the fourth line follows from translation invariance of the L^p norms. Thus by Proposition 5.4, T extends to a unique operator on all of $L^1(\mathbb{R})$ satisfying the same estimate. Now define the operator T on $L^{p_1}(\mathbb{R})$, where $p_1 = \infty$, initially on just the set of all simple functions by $Tf := (f * \mathcal{P}_y)(x)$. As before, the proof is trivial if $y = 0$. So assume $y \neq 0$. Then for each x ,

$$\begin{aligned} |Tf(x)| &= \left| \int_{\mathbb{R}} \frac{1}{\pi} \frac{f(x-t)y}{y^2 + t^2} dt \right| \\ &\leq \int_{\mathbb{R}} |f(x-t)| \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &\leq \int_{\mathbb{R}} \|f(x-\cdot)\|_{L^\infty(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt \\ &= \int_{\mathbb{R}} \|f\|_{L^\infty(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{y^2 + t^2} \right| dt = \|f\|_{L^\infty(\mathbb{R})} \cdot \|\mathcal{P}_y\|_{L^1(\mathbb{R})} = \|f\|_{L^\infty(\mathbb{R})}. \end{aligned}$$

Using Proposition 5.4, the result extends to all of $L^\infty(\mathbb{R})$. On the Riesz diagram, we have proven the estimate for the endpoints $(0,0)$ and $(1,1)$. Thus by the Riesz-Thorin interpolation theorem, the theorem follows for all points on the diagonal line connecting these endpoints (i.e., for all $1 \leq p \leq \infty$).

(b) Now let $1 \leq p < \infty$, and let $\delta > 0$ be fixed. By continuity of translations in $L^p(\mathbb{R})$, there exists $\varepsilon > 0$

such that $|t| \leq \delta \implies \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} < \varepsilon/2$. Thus we compute,

$$\begin{aligned} \|f * \mathcal{P}_y - f\|_{L^p} &= \left(\int_{\mathbb{R}} \left| \int_{\mathbb{R}} \frac{1}{\pi} \frac{f(x-t)y}{t^2 + y^2} dt - f(x) \right|^p dx \right)^{1/p} \\ &= \left(\int_{\mathbb{R}} \left| \int_{\mathbb{R}} [f(x-t) - f(x)] \cdot \frac{1}{\pi} \frac{y}{t^2 + y^2} dt \right|^p dx \right)^{1/p} \\ &\leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x-t) - f(x)| \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right|^p dx \right)^{1/p} dt \\ &= \int_{\mathbb{R}} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt \\ &= \int_{|t| \leq \delta} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt + \int_{|t| > \delta} \|f(\cdot - t) - f\|_{L^p(\mathbb{R})} \cdot \left| \frac{1}{\pi} \frac{y}{t^2 + y^2} \right| dt \\ &\leq \varepsilon/2 + 2 \|f\|_{L^p(\mathbb{R})} \cdot \int_{|t| > \delta} |\mathcal{P}_y(t)| dt, \end{aligned}$$

where the second line follows from $\|\mathcal{P}_y\|_{L^1(\mathbb{R})} = 1$ for any $y \in \mathbb{R}$, and the third line follows from Minkowski's inequality. Since $\mathcal{P}_y(t) \rightarrow 0$ for all $t \neq 0$ in the limit $y \rightarrow 0^+$, it follows that the second term tends to 0 in the limit $y \rightarrow 0^+$. I.e., there exists some y_0 so that for all $y < y_0$, the second term is bounded above by $\varepsilon/(2\|f\|_{L^p(\mathbb{R})})$. Therefore, we conclude that $\|f * \mathcal{P}_{y_0} - f\|_{L^p(\mathbb{R})} \leq \varepsilon$ so that $f * \mathcal{P}_y \rightarrow f$ in the L^p norm.

Problem 2.16. Prove that the space $\mathbf{H}_r^1$ is complete as follows. Let $\{f_n\}$ be a Cauchy sequence in $\mathbf{H}_r^1$. Then since $\{f_n\}$ is also Cauchy in L^1 , there is an L^1 function f so that $f = \lim_{n \rightarrow \infty} f_n$ in the L^1 -norm. Now for an appropriate subsequence $\{n_k\}$, write $f = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k})$.

Let $\mathbf{H}_r^1$ denote the Hardy space, and let $\{f_n\} \subset \mathbf{H}_r^1$ be a Cauchy sequence. First, we begin by showing that each $f \in L^1$. Let $\sum_j \lambda_j \mathbf{a}_j$ be an arbitrary decomposition of f , where each $\mathbf{a}_j$ is an atom, and $\sum |\lambda_j| < \infty$. Thus since the sum converges under the L^1 norm, Fubini/Tonelli implies that

$$\int_{\mathbb{R}} |f| dx = \int_{\mathbb{R}} \left| \sum_1^{\infty} \lambda_k \mathbf{a}_k \right| \leq \int_{\mathbb{R}} \sum_1^{\infty} |\lambda_k| |\mathbf{a}_k| = \sum_1^{\infty} \int_{\mathbb{R}} |\lambda_k| |\mathbf{a}_k| = \sum_1^{\infty} |\lambda_k| < \infty.$$

Thus since $\sum \lambda_j \mathbf{a}_j$ was an arbitrary decomposition of f into a sum of atoms, it follows that $f \in L^1$. In particular, taking the infimum over all such decompositions also yields the estimate $\|f\|_{L^1} \leq \|f\|_{\mathbf{H}_r^1}$. Thus if $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, they are also Cauchy in L^1 . Since the latter is a Banach space, there exists an L^1 function f such that $f = \lim_{n \rightarrow \infty} f_n$ in the L^1 -norm. But this implies there exists a subsequence $\{f_{n_k}\}$ that converges to f pointwise a.e. Fix $\varepsilon > 0$. Since $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, for every $k \geq 1$, there exists a positive integer N_k such that for all $m, n > N_k$,

$$\|f_m - f_n\|_{\mathbf{H}_r^1} < \frac{\varepsilon}{2^k}.$$

Thus from the subsequence $\{f_{n_k}\}_{k=1}^{\infty}$, we construct another subsequence as follows: we choose $n_1 > N_1$. Next, we choose $n_2 > \max\{n_1, N_2\}$. Repeating inductively, we obtain a subsequence $\{f_{n_k}\}$ such that

$$\sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}_r^1} \leq \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon < \infty.$$

Now we recall that

$$f = \lim_{j \rightarrow \infty} f_{n_j} = f_{n_1} + \lim_{j \rightarrow \infty} \sum_{k=1}^j (f_{n_{k+1}} - f_{n_k}) = f_{n_1} + \sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k}).$$

Thus

$$\|f\|_{\mathbf{H}_r^1} \leq \|f_{n_1}\|_{\mathbf{H}_r^1} + \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}_r^1} < \infty,$$

which establishes that $f \in \mathbf{H}_r^1$. All that remains to be shown is that $f_n \rightarrow f$ under the $\mathbf{H}_r^1$ norm. We compute,

$$\|f - f_n\|_{\mathbf{H}_r^1} \leq \|f - f_{n_k}\|_{\mathbf{H}_r^1} + \|f_{n_k} - f_n\|_{\mathbf{H}_r^1}.$$

Since the sequence $\{f_n\}$ is Cauchy in $\mathbf{H}_r^1$, the second term vanishes in the limit $n, k \rightarrow \infty$. On the other hand, since

$$\|f - f_{n_k}\|_{\mathbf{H}_r^1} \leq \sum_{j=k}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_{\mathbf{H}_r^1}.$$

By our construction of the subsequence $\{f_{n_k}\}$, we can control the sum by ε . Thus we conclude that $f_n \rightarrow f$ under the $\mathbf{H}_r^1$ norm, which means that $\mathbf{H}_r^1$ is a Banach space.

Problem 2.20. When $f \in L^1(\mathbb{R}^d)$ we know that its Fourier transform is bounded and $\hat{f}(\xi)$ tends to 0 as $|\xi| \rightarrow \infty$ (the Riemann-Lebesgue lemma), but no better assertion about the “smallness” of $\hat{f}$ can be made. (For the analogous result for Fourier series, see Chapter 3 in Book III.) Show, however, that for $f \in \mathbf{H}_r^1$ we have

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \leq A \|f\|_{\mathbf{H}_r^1}.$$

[Hint: Verify this for atoms.]

First, assume that for every atom $\mathbf{a}$, the above estimate holds. Then for any decomposition, $f = \sum_{k=1}^{\infty} \lambda_k \mathbf{a}_k$ into atoms,

$$\begin{aligned} \int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} &= \int_{\mathbb{R}^d} \left| \sum_1^{\infty} \lambda_k \hat{\mathbf{a}}_k \right| \frac{d\xi}{|\xi|^d} \\ &\leq \sum_1^{\infty} \int_{\mathbb{R}^d} |\lambda_k| |\hat{\mathbf{a}}_k| \frac{d\xi}{|\xi|^d} \\ &\leq A \sum_1^{\infty} |\lambda_k|. \end{aligned}$$

Thus taking the infimum over all such decompositions of f , we obtain,

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \lesssim \|f\|_{\mathbf{H}_r^1}.$$

So now, we will show that the result holds true for every atom. Let $\mathbf{a}$ be an atom on $\mathbb{R}^d$; without loss of generality, we can assume that $\mathbf{a}$ is supported on the unit ball $B_1(0) \subset \mathbb{R}^d$. Assume $\mathbf{a}$ is not identically zero. Then one has that

$$\hat{\mathbf{a}}(\xi) := \int_{\mathbb{R}^d} \mathbf{a}(x) e^{-2\pi i x \cdot \xi} dx = \int_{|x| \leq 1} \mathbf{a}(x) e^{-2\pi i x \cdot \xi} dx = \int_{|x| \leq 1} \mathbf{a}(x) (e^{-2\pi i x \cdot \xi} - 1) dx,$$

where the final equality follows from the fact that $\int_{\mathbb{R}^d} \mathbf{a}(x) dx = 0$. Thus,

$$\begin{aligned} \int_{\mathbb{R}^d} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} &= \int_{|\xi| \leq 1} \left| \int_{|x| \leq 1} \mathbf{a}(x) \frac{e^{-2\pi i x \cdot \xi} - 1}{|\xi|^d} dx \right| d\xi + \int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} \\ &\leq \int_{|x| \leq 1} \int_{|\xi| \leq 1} |\mathbf{a}(x)| \cdot \left| \frac{e^{-2\pi i x \cdot \xi} - 1}{|\xi|^d} \right| d\xi dx + \int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)| \frac{d\xi}{|\xi|^d} \\ &\leq 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\int_{|\xi| > 1} |\hat{\mathbf{a}}(\xi)|^2 d\xi \right)^{1/2} \left(\int_{|\xi| > 1} \frac{1}{|\xi|^{2d}} d\xi \right)^{1/2} \\ &= 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\int_{|x| \leq 1} |\mathbf{a}(x)|^2 dx \right)^{1/2} \\ &\leq 2\pi \|\mathbf{a}\|_{L^1(\mathbb{R}^d)} + \left(\frac{1}{m(B_1(0))} \right)^{1/2} \leq 2\pi + \frac{1}{m(B_1(0))^{1/2}} \equiv A < \infty. \end{aligned}$$

Problem 3.1. Suppose F is a distribution on Ω and $F = f$, with f a C^k function in Ω . Show that $\partial_x^\alpha F$, taken in the sense of distributions, agrees with $\partial_x^\alpha f$ for each $|\alpha| \leq k$.

Let $\Omega \subseteq \mathbb{R}^d$, and let F be a distribution on Ω such that $F = f$, where $f \in C^k(\Omega)$. Recall that we defined $\partial_x^\alpha F$ as the distribution,

$$(\partial_x^\alpha F)(\varphi) := (-1)^{|\alpha|} F(\partial_x^\alpha \varphi), \quad \forall \varphi \in C_c^\infty(\Omega).$$

When $|\alpha| = 1$, we compute,

$$\begin{aligned} \int_{\Omega} (\partial_{x_j} f) \varphi \, dx &= f \varphi \Big|_{\partial\Omega} - \int_{\Omega} f \partial_{x_j} \varphi \, dx \\ &= - \int_{\Omega} f \partial_{x_j} \varphi \, dx = -F(\partial_{x_j} \varphi), \end{aligned}$$

where the boundary term is zero since $\text{supp}(\varphi) \subset \Omega$. Now for each $1 \leq |\alpha| \leq k$, applying induction on the above steps (by “swapping” one derivative at a time), we obtain,

$$\int_{\Omega} (\partial_x^\alpha f) \varphi \, dx = (-1)^{|\alpha|} \int_{\Omega} f \partial_x^\alpha \varphi \, dx = (-1)^{|\alpha|} F(\partial_x^\alpha \varphi),$$

for all $\varphi \in C_c^\infty(\Omega)$.

Problem 3.11. Suppose that $\varphi_1, \varphi_2 \in \mathcal{S}$.

- (a) Verify that $\varphi_1 \cdot \varphi_2$ belongs to $\mathcal{S}$.
- (b) Using the Fourier transform, prove that $\varphi_1 * \varphi_2 \in \mathcal{S}$.
- (c) Show directly from the definition of convolution that $\varphi_1 * \varphi_2 \in \mathcal{S}$.

Let $\varphi_1, \varphi_2 \in \mathcal{S}$.

- (a) Let N be an arbitrary fixed positive integer. For any $x \in \mathbb{R}^d$, and $|\alpha|, |\beta| \leq N$, one has that

$$x^\beta \partial_x^\alpha (\varphi_1 \cdot \varphi_2)(x) = x^\beta \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} (\partial_x^\gamma \varphi_1)(\partial_x^{\alpha-\gamma} \varphi_2).$$

This means that

$$\begin{aligned} |x^\beta \partial_x^\alpha (\varphi_1 \cdot \varphi_2)(x)| &\leq \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} |x^\beta (\partial_x^\gamma \varphi_1)(x) (\partial_x^{\alpha-\gamma} \varphi_2)(x)| \\ &\leq \sum_{\gamma \leq \alpha} \binom{\alpha}{\gamma} |x^\beta (\partial_x^\gamma \varphi_1)(x)| \|\partial_x^{\alpha-\gamma} \varphi_2\|_{L^\infty(\mathbb{R}^d)}. \end{aligned}$$

Thus taking the supremum over all $x \in \mathbb{R}^d$ and $|\alpha|, |\beta| \leq N$ gives $\|\varphi_1 \cdot \varphi_2\|_N < \infty$. Since this is true for all positive integers N , the proof concludes.

- (b) Since $\varphi_{1,2}$ are Schwartz, both of their Fourier transforms are also Schwartz. Thus by (a), $\hat{\varphi}_1 \cdot \hat{\varphi}_2$ is also Schwartz. But $\hat{\varphi}_1 \cdot \hat{\varphi}_2 = (\varphi_1 * \varphi_2)^\wedge$. Therefore, $(\varphi_1 * \varphi_2)^\wedge \in \mathcal{S}$. Since the inverse Fourier transform also takes Schwartz functions to Schwartz functions, we conclude that $\varphi_1 * \varphi_2$ is Schwartz.

- (c) Recall that

$$(\varphi_1 * \varphi_2)(x) = \int_{\mathbb{R}^d} \varphi_1(x-y) \varphi_2(y) \, dy.$$

Therefore, for any $x \in \mathbb{R}^d$, fixed positive integer N , and $|\alpha|, |\beta| \leq N$,

$$|x^\beta \partial_x^\alpha (\varphi_1 * \varphi_2)(x)| = \left| x^\beta \int_{\mathbb{R}^d} \partial_x^\alpha \varphi_1(x-y) \varphi_2(y) \, dy \right| \leq \int_{\mathbb{R}^d} |x^\beta \partial_x^\alpha \varphi_1(x-y)| |\varphi_2(y)| \, dy$$

2.2 Evans

Problem 2.18. (Stokes' rule) Assume u solves the initial-value problem

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = 0, u_t = h, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Show that $v := u_t$ solves

$$\begin{cases} v_{tt} - \Delta v = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ v = h, v_t = 0, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

This is *Stokes' rule*.

Assume u solves the given IVP, and let $v := u_t$. One immediately notes that on $\mathbb{R}^n \times \{t = 0\}$, since $u_t = h$, $v = h$. Second, $v_t = u_{tt}$. On $\mathbb{R}^n \times \{t = 0\}$, $u \equiv 0$ so that $\Delta u = 0$. But since u satisfies the wave equation $u_{tt} = \Delta u \equiv 0$. Thus $v_t \equiv 0$ on $\mathbb{R}^n \times \{t = 0\}$. Next we note that ∂_t and ∂_{x_j} commute for any $1 \leq j \leq n$. Thus since u satisfies $u_{tt} - \Delta u = 0$, differentiating both sides by t , one obtains

$$u_{ttt} - \Delta u_t = 0 \iff v_{tt} - \Delta v = 0$$

in $\mathbb{R}^n \times (0, \infty)$.

Problem 2.19.

(a) Show the general solution of the PDE $u_{xy} = 0$ is

$$u(x, y) = F(x) + G(y)$$

for arbitrary functions F, G .

(b) Using the change of variables $\xi = x + t$, $\eta = x - t$, show $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$.

(c) Use (a) and (b) to rederive d'Alembert's formula.

(d) Under what conditions on the initial data g, h is the solution u a right-moving wave? A left-moving wave?

(a) Consider the PDE $u_{xy} = 0$. Let $v(x, y) = u_x$. Then v satisfies the differential equation $v_y = 0$. Thus since v is independent of y , one has $u_x(x, y) = v(x, y) = f(x)$ for some arbitrary function f . But then integrating both sides with respect to x , and supposing that $F(x)$ is the antiderivative of $f(x)$,

$$u(x, y) = F(x) + G(y),$$

where G is some arbitrary function of y .

(b) Define $\xi = x + t$ and $\eta = x - t$ so that $x = (\xi + \eta)/2$ and $t = (\xi - \eta)/2$. Then,

$$\begin{aligned} u_\xi &= u_x \frac{\partial x}{\partial \xi} + u_t \frac{\partial t}{\partial \xi} \\ &= \frac{1}{2} u_x + \frac{1}{2} u_t. \\ u_{\xi\eta} &= \frac{1}{2} \left[u_{xx} \frac{\partial x}{\partial \eta} + u_{xt} \frac{\partial t}{\partial \eta} \right] + \frac{1}{2} \left[u_{tx} \frac{\partial x}{\partial \eta} + u_{tt} \frac{\partial t}{\partial \eta} \right] \\ &= \frac{1}{2} [u_{xx} - u_{xt}] + \frac{1}{2} [u_{tx} - u_{tt}] = \frac{1}{2} [u_{tt} - u_{xx}], \end{aligned}$$

where we used the fact that $u_{xt} = u_{tx}$. Thus, the proof concludes.

(c) Consider the initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Define the change of variables $\xi = x + t$ and $\eta = x - t$ as above. By (b), $u_{tt} - u_{xx} = 0$ if and only if $u_{\xi\eta} = 0$. By (a), it follows that u must be of the form,

$$u(\xi, \eta) = F(\xi) + G(\eta).$$

Thus, $u(x, t) = F(x + t) + G(x - t)$. Now we must match the initial conditions. At $t = 0$,

$$u(x, 0) = F(x) + G(x) = g(x).$$

$$u_t(x, 0) = F'(x) - G'(x) = h(x).$$

Differentiating the first equation, one first obtains the system

$$\begin{cases} F'(x) + G'(x) = g'(x) \\ F'(x) - G'(x) = h(x). \end{cases}$$

Thus, we get

$$F'(x) = \frac{1}{2}(g'(x) + h(x)) \implies F(x + t) = \frac{1}{2}[g(x + t) - g(0)] + \frac{1}{2} \int_0^{x+t} h(y) dy + F(0).$$

Likewise, we get

$$G(x - t) = \frac{1}{2}[g(x - t) - g(0)] - \int_0^{x-t} h(y) dy + G(0).$$

Adding the two equations, and using the fact that $F(0) + G(0) = g(0)$, we conclude that

$$u(x, t) = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy,$$

which is precisely d'Alembert's formula.

- (d) To get a left-moving wave, we want the functions of $(x - t)$ to cancel out. Suppose $h(x) = g'(x)$. Then the fundamental theorem of calculus implies that

$$u(x, t) = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2} \int_{x-t}^{x+t} g'(y) dy = \frac{1}{2}[g(x - t) + g(x + t)] + \frac{1}{2}[g(x + t) - g(x - t)] = g(x + t).$$

Likewise, to get a right-moving wave, we want the functions of $(x + t)$ to cancel out, so we can take $h(x) = -g'(x)$.

Problem 2.21. (Equipartition of Energy). Let $u \in C^2(\mathbb{R} \times [0, \infty))$ solve the initial-value problem for the wave equation in one-dimension:

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R} \times (0, \infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R} \times \{t = 0\}). \end{cases}$$

Suppose g, h have compact support. The *kinetic energy* is $k(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_t^2(x, t) dx$ and the *potential energy* is $p(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_x^2(x, t) dx$. Prove

- (i) $k(t) + p(t)$ is constant in t ,
(ii) $k(t) = p(t)$ for all large enough times t .

- (i) Let us define the kinetic and potential energy, $k(t)$ and $p(t)$ as given above. We wish to show that the total energy

$$e(t) = k(t) + p(t) = \frac{1}{2} \int_{-\infty}^{\infty} u_t^2 + u_x^2 dx$$

is conserved, where u is given by d'Alembert's formula,

$$u(x, t) = \frac{1}{2}[g(x + t) + g(x - t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy.$$

so that

$$\begin{aligned} u_t(x, t) &= \frac{1}{2}[g'(x+t) - g'(x-t) + h(x+t) + h(x-t)], \\ u_x(x, t) &= \frac{1}{2}[g'(x+t) + g'(x-t) + h(x+t) - h(x-t)]. \end{aligned}$$

Let $T > 0$, and let $K \subset \mathbb{R}$ be a compact set large enough so that $\text{supp } g, \text{supp } h$ are both contained in K . Then it follows that u_t^2 and u_x^2 are both supported in $K_T := K + [-T, T]$ for all $0 \leq t \leq T$. Since u_t^2 and u_x^2 are continuous on $\mathbb{R}$ (and thus on K_T) for all $0 \leq t \leq T$, by the extreme value theorem, there exist finite constants M and N so that for all $x \in K$ and $0 \leq t \leq T$, $|u_t|^2 \leq M\chi_{K_T}$ and $|u_x|^2 \leq N\chi_{K_T}$, where both dominating functions are integrable since K_T is compact. Thus using the Dominated Convergence Theorem, taking the derivative of $e(t)$ gives us,

$$\begin{aligned} \dot{e}(t) &= \frac{1}{2} \frac{\partial}{\partial t} \int_{-\infty}^{\infty} u_t^2 + u_x^2 dx = \frac{1}{2} \frac{\partial}{\partial t} \int_{K_T} u_t^2 + u_x^2 dx \\ &= \frac{1}{2} \int_{K_T} \partial_t(u_t^2 + u_x^2) dx \\ &= \int_{K_T} u_t u_{tt} + u_x \cdot \partial_x u_t dx \\ &= \int_{K_T} u_t(u_{tt} - u_{xx}) dx \quad (\text{by integration by parts}) \\ &= 0, \end{aligned}$$

where there is no boundary term since u_x, u_t are supported inside K_T . So we have shown that $\dot{e}(t) = 0$ for all $0 \leq t \leq T$. Since $T > 0$ was chosen arbitrarily, it follows that $\dot{e}(t) \equiv 0$. Therefore, the total energy is constant in t .

- (ii) What we wish is that in the limit $t \rightarrow \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| \rightarrow 0.$$

First, we compute

$$u_t^2 - u_x^2 = h(x+t)h(x-t) - h(x+t)g'(x-t) + h(x-t)g'(x+t) - g'(x-t)g'(x+t).$$

Then

$$\begin{aligned} \left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| &\leq \int_{-\infty}^{\infty} |h(y)h(y-2t)| dy + \int_{-\infty}^{\infty} |h(y)g'(y-2t)| dy \\ &\quad + \int_{-\infty}^{\infty} |h(y-2t)g'(y)| dy + \int_{-\infty}^{\infty} |g'(y-2t)g'(y)| dy. \end{aligned}$$

Let $K \subset \mathbb{R}$ be a compact set large enough that contains the support of both g and h . Then we may write,

$$\begin{aligned} \left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| &\leq \int_K |h(y)h(y-2t)| dy + \int_K |h(y)g'(y-2t)| dy \\ &\quad + \int_K |h(y-2t)g'(y)| dy + \int_K |g'(y-2t)g'(y)| dy. \end{aligned}$$

We will only show that the first term on the right vanishes since the work for the remaining terms is identical. Since $h(u) \rightarrow 0$ as $|u| \rightarrow \infty$ (by compact support of h), we know that pointwise $|h(y)h(y-2t)| \rightarrow 0$ as $t \rightarrow \infty$. Next since h is continuous, there exists some finite $M > 0$ so that $|h(y)|, |h(y-2t)| \leq M$ for all $y \in K$. Thus, $|h(y)h(y-2t)| \leq M^2\chi_K \in L^1(\mathbb{R})$ since K is compact. Hence, by the Dominated Convergence Theorem, one has that the first term vanishes as $t \rightarrow \infty$. As we stated before, the remaining three terms also vanish as $t \rightarrow \infty$. Thus, we conclude that as $t \rightarrow \infty$,

$$\left| \int_{\mathbb{R}} u_t^2 - u_x^2 dx \right| \rightarrow 0.$$

2.3 Luk Notes

Problem N.1. (Luk, pg. 3) The Einstein vacuum equations describe the propagation of gravitational waves in the absence of matter and take the form $\text{Ric}(g) = 0$, where the Lorentzian metric g is the unknown. Show that in a coordinate system, these equations take the form

$$\text{Ric}(g)_{\mu\nu} = -\frac{1}{2}g^{\alpha\beta}\partial_{\alpha\beta}^2 g_{\mu\nu} - \frac{1}{2}g^{\alpha\beta}\partial_{\mu\nu}^2 g_{\alpha\beta} + \frac{1}{2}g^{\alpha\beta}\partial_{\alpha\nu}^2 g_{\beta\mu} + \frac{1}{2}g^{\alpha\beta}\partial_{\beta\mu}^2 g_{\alpha\nu} + F_{\mu\nu}(g, \partial g),$$

where $F_{\mu\nu}(g, \partial g)$ is a function of g and its derivatives.

Let (M, g) be a Lorentzian manifold, and let $\text{Ric}(g)$ denote the Ricci curvature tensor on M . Let $(x^j)_{j=1}^n$ denote a local coordinate system, and let $(\partial_j)_{j=1}^n$ be the corresponding vector fields. Let R denote the Riemann curvature tensor. Then if R is given in coordinates by $R = R_{ijk}{}^\ell dx^i \otimes dx^j \otimes dx^k \otimes \partial_\ell$, then

$$R_{ijk}{}^\ell = \partial_i \Gamma_{jk}^\ell - \partial_j \Gamma_{ik}^\ell + \Gamma_{jk}^m \Gamma_{im}^\ell - \Gamma_{ik}^m \Gamma_{jm}^\ell,$$

where the $\Gamma_{**}^\bullet$ are the *Christoffel symbols*. One can also show that in coordinates,

$$\Gamma_{jk}^i = \frac{1}{2}g^{im}(\partial_k g_{mj} + \partial_j g_{mk} - \partial_m g_{jk}).$$

Thus, we then compute,

$$R_{ijk}{}^\ell = \frac{1}{2}g^{\ell m}(\partial_{ki}^2 g_{mj} + \partial_{ji}^2 g_{mk} - \partial_{mi}^2 g_{jk}) + \{\dots\} - \frac{1}{2}g^{\ell m}(\partial_{kj}^2 g_{mi} + \partial_{ij}^2 g_{mk} - \partial_{mj}^2 g_{ik}) - \{\dots\} + \{\dots\}.$$

Then the component functions of the Ricci curvature tensor in coordinates are,

$$\text{Ric}(g)_{ij} = R_{\ell ij}{}^\ell = -\frac{1}{2}g^{\ell m}\partial_{\ell m}^2 g_{ij} - \frac{1}{2}g^{\ell m}\partial_{ij}^2 g_{\ell m} + \frac{1}{2}g^{\ell m}\partial_{im}^2 g_{\ell j} + \frac{1}{2}g^{\ell m}\partial_{\ell j}^2 g_{im} + F_{ij}(g, \partial g).$$

Thus to agree with the format presented in the problem, let $\ell = \alpha$, $m = \beta$, $i = \mu$, and $j = \nu$.

Problem N.2. (Luk, pg. 18) Let ϕ be a solution to the Cauchy problem,

$$\begin{cases} \partial_\alpha(a^{\alpha\beta}\partial_\beta\phi) = F \\ (\phi, \partial_t\phi)|_{\{t=0\}} = (\phi_0, \phi_1). \end{cases}$$

Show that

$$\sup_{t \in [0, T]} \|\phi\|_{L^2(\mathbb{R}^n)}(t) \leq C \left(\|\phi_0\|_{L^2(\mathbb{R}^n)} + \int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)}(t) dt \right).$$

Let ϕ be a sufficiently regular solution to the aforementioned Cauchy problem. Then the fundamental theorem of calculus tells us that

$$\phi(x, t) = \phi_0(x) + \int_0^t \partial_t\phi(t', x) dt',$$

for any $t \in [0, T]$. Thus, we compute,

$$\begin{aligned} |\phi(x, t)|^2 &\leq |\phi_0|^2 + \left| \int_0^t \partial_t\phi(t', x) dt' \right|^2 \\ \implies \|\phi\|_{L^2(\mathbb{R}^n)}^2 &\leq 2\|\phi_0\|_{L^2(\mathbb{R}^n)}^2 + 2 \int_{\mathbb{R}^n} \left| \int_0^t \partial_t\phi dt' \right|^2 dx \\ \implies \|\phi\|_{L^2(\mathbb{R}^n)} &\leq 2^{1/2} \|\phi_0\|_{L^2(\mathbb{R}^n)} + 2^{1/2} \left(\int_{\mathbb{R}^n} \left| \int_0^t \partial_t\phi dt' \right|^2 dx \right)^{1/2} \\ &\leq 2^{1/2} \|\phi_0\|_{L^2(\mathbb{R}^n)} + 2^{1/2} \int_0^t \|\partial_t\phi\|_{L^2(\mathbb{R}^n)} dt' \\ &\leq 2^{1/2} \left(\|\phi_0\|_{L^2(\mathbb{R}^n)} + \int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)} dt \right), \end{aligned}$$

where the fourth line follows the Minkowski inequality, and the final line follows from the fact that $\|\partial_t \phi\|_{L^2(\mathbb{R}^n)} \geq 0$.

Problem N.3. Consider the general functional inequality,

$$\|\eta\|_{H^s(\mathbb{R}^n)} \leq C \|\eta\|_{H^{s_1}(\mathbb{R}^n)}^{\theta_1} \|\eta\|_{H^{s_2}(\mathbb{R}^n)}^{\theta_2}$$

for some $C = C(s_1, s_2, s, n) > 0$, where $0 \leq s_1 \leq s \leq s_2$, $\theta_1 + \theta_2 = 1$, and $\theta_1 s_1 + \theta_2 s_2 = s$. Prove this functional inequality by taking the Fourier transform and using Hölder's inequality.

Let $\eta \in H^s(\mathbb{R}^n)$. We recall that

$$\|\eta\|_{H^s(\mathbb{R}^n)}^2 := \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{\eta}|^2 d\xi$$

Let $0 \leq s_1 \leq s \leq s_2$, and let θ_1, θ_2 such that $\theta_1 + \theta_2 = 1$ and $\theta_1 s_1 + \theta_2 s_2 = s$. Then we can write,

$$\begin{aligned} \|\eta\|_{H^s(\mathbb{R}^n)}^2 &= \int_{\mathbb{R}^n} (1 + |\xi|^2)^{\theta_1 s_1} |\hat{\eta}|^{2\theta_1} (1 + |\xi|^2)^{\theta_2 s_2} |\hat{\eta}|^{2\theta_2} d\xi \\ &\leq \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^{s_1} |\hat{\eta}|^2 d\xi \right)^{\theta_1} \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^{s_2} |\hat{\eta}|^2 d\xi \right)^{\theta_2} \\ &= \|\eta\|_{H^{s_1}(\mathbb{R}^n)}^{2\theta_1} \|\eta\|_{H^{s_2}(\mathbb{R}^n)}^{2\theta_2}. \end{aligned}$$

Problem E.1.5. Given smooth and compactly supported initial data to the linear wave equation in any dimension, use the Fourier representation formula to show that the $\dot{H}^k$ norm is uniformly bounded in time for any $k \geq 1$.

Let $(\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1)$ be smooth and compactly supported initial data to the linear wave equation in $\mathbb{R}^{1+n}$, and let ϕ be the unique solution to the Cauchy wave problem. In the notes, we explicitly showed that

$$\hat{\phi}(t, \xi) = \hat{\phi}_0 \cos(2\pi t|\xi|) + \frac{\hat{\phi}_1}{2\pi|\xi|} \sin(2\pi t|\xi|).$$

We also recall that the norms for the spaces $\dot{H}^k$ (i.e., the homogeneous Sobolev spaces $\dot{W}^{k,2}$) for $1 \leq k < \infty$ are given by

$$\|u\|_{\dot{H}^k(\mathbb{R}^n)} := \left(\int_{\mathbb{R}^n} |\xi|^{2k} |\hat{u}|^2 d\xi \right)^{1/2}.$$

Thus, we compute,

$$\begin{aligned} \|\phi\|_{\dot{H}^k(\mathbb{R}^n)}^2(t) &= \int_{\mathbb{R}^n} |\xi|^{2k} \left| \hat{\phi}_0 \cos(2\pi t|\xi|) + \frac{\hat{\phi}_1}{2\pi|\xi|} \sin(2\pi t|\xi|) \right|^2 d\xi \\ &\leq 2 \int_{\mathbb{R}^n} |\xi|^{2k} |\hat{\phi}_0|^2 \cos^2(2\pi t|\xi|) d\xi + \frac{1}{2\pi^2} \int_{\mathbb{R}^n} |\xi|^{2k-2} |\hat{\phi}_1|^2 \sin^2(2\pi t|\xi|) d\xi \\ &\leq 2 \|\phi_0\|_{\dot{H}^k(\mathbb{R}^n)}^2 + \frac{1}{2\pi^2} \|\phi_1\|_{\dot{H}^{k-1}(\mathbb{R}^n)}^2, \end{aligned}$$

for all $t \geq 0$. Thus, we conclude that the $\dot{H}^k$ is uniformly bounded in time for any $1 \leq k < \infty$.

Problem E.1.6. (Decay for Klein-Gordon equation) Suppose ϕ satisfies the Klein-Gordon equation; i.e.,

$$\square\phi - \phi = 0$$

on $\mathbb{R} \times \mathbb{R}^n$ with initial data given by $(\phi, \partial_t \phi)|_{t=0} = (\phi_0, \phi_1) \in C_c^\infty \times C_c^\infty$.

- (a) First, show using the Fourier transform that if the solution is sufficiently regular, it is given by the formula

$$\hat{\phi}(t, \xi) = \hat{\phi}_0(\xi) \cos(\sqrt{4\pi^2|\xi|^2 + 1}t) + \frac{\hat{\phi}_1(\xi)}{\sqrt{4\pi^2|\xi|^2 + 1}} \sin(\sqrt{4\pi^2|\xi|^2 + 1}t).$$

- (b) For the $n = 1$ case, prove that $(\sup_x |\phi|)(t) \leq \frac{C}{(1+t)^{\frac{1}{2}}}$, where $C = C(\phi_0, \phi_1) > 0$ is independent of t .

Note that it suffices (why?) to show that

$$\left| \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x \cdot \xi}{t} + \sqrt{4\pi^2\xi^2 + 1}\right)\right) \hat{\phi}_0(\xi); d\xi \right| \leq \frac{C}{t^{1/2}}$$

for t large. Let

$$\varphi := \frac{2\pi x \cdot \xi}{t} + \sqrt{4\pi^2|\xi|^2 + 1}.$$

Show that

$$\frac{\partial}{\partial \xi} \varphi = 0 \implies \xi^2 = \frac{x^2}{4\pi^2(t^2 - x^2)}.$$

Show also that

$$\frac{\partial^2}{\partial \xi^2} \varphi \geq \frac{4\pi^2}{(4\pi^2\xi^2 + 1)^{\frac{3}{2}}}.$$

Now split the integral into $\int_{|\xi^2 - \frac{x^2}{4\pi^2(t^2 - x^2)}| \leq \delta} + \int_{|\xi^2 - \frac{x^2}{4\pi^2(t^2 - x^2)}| \geq \delta}$, estimate each part with the above observations and optimized in δ to obtain the desired decay result.

- (c) Finally, can you prove that

$$|\phi| \leq \frac{C}{(1+t)^{\frac{n}{2}}}$$

holds for some $C = C(\phi_0, \phi_1) > 0$ for arbitrary dimensions?

- (a) Consider the Klein-Gordon equation $\square\phi - \phi = 0$ on $\mathbb{R} \times \mathbb{R}^n$, with compactly supported smooth initial data. Assume that a solution exists and that it is sufficiently regular. Then taking the Fourier transform of the equation, one obtains

$$-\partial_t^2 \hat{\phi} - (4\pi^2|\xi|^2 + 1)\hat{\phi} = 0 \implies \partial_t^2 \hat{\phi} = (4\pi^2|\xi|^2 + 1)\hat{\phi}.$$

Thus solving the ODE and using the initial data, we obtain (as desired)

$$\hat{\phi}(t, \xi) = \hat{\phi}_0 \cos(\sqrt{4\pi^2|\xi|^2 + 1}t) + \frac{\hat{\phi}_1}{\sqrt{4\pi^2|\xi|^2 + 1}} \sin(\sqrt{4\pi^2|\xi|^2 + 1}t).$$

- (b) Rewriting the formula for $\hat{\phi}$ (from (a)) in a more convenient way, we obtain

$$\hat{\phi}(t, \xi) = \exp\left(i\sqrt{4\pi^2\xi^2 + 1}t\right) \left(\frac{\hat{\phi}_0}{2} + \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2 + 1}}\right) + \exp\left(-i\sqrt{4\pi^2\xi^2 + 1}t\right) \left(\frac{\hat{\phi}_0}{2} - \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2 + 1}}\right).$$

Thus taking the inverse Fourier transform,

$$\phi(t, x) = I_1 + I_2 + I_3 + I_4,$$

where

$$\begin{aligned} I_1 &= \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2 + 1}\right)\right) \frac{\hat{\phi}_0}{2} d\xi & I_2 &= \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2 + 1}\right)\right) \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2 + 1}} d\xi \\ I_3 &= \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} - \sqrt{4\pi^2\xi^2 + 1}\right)\right) \frac{\hat{\phi}_0}{2} d\xi & I_4 &= \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x\xi}{t} - \sqrt{4\pi^2\xi^2 + 1}\right)\right) \frac{\hat{\phi}_1}{2i\sqrt{4\pi^2\xi^2 + 1}} d\xi. \end{aligned}$$

As suggested in the problem, it suffices to consider just $t \geq 1$. Since ϕ_0, ϕ_1 are compactly supported smooth functions, their Fourier transforms are Schwartz (thus integrable). This means there exist $M_j > 0$, for $j = 1, 2, 3, 4$ such that $|I_j| \leq M_j$. Thus, taking $M > M_1 + \dots + M_4$, we have that for all $x \in \mathbb{R}$, $|\phi| \leq M$. Taking $C > 0$ large enough so that $C \geq M \cdot 2^{1/2}$, we have $|\phi(t, x)| \leq C/(1+t)^{1/2}$ for all $t \geq 0$ and $x \in \mathbb{R}$ so that the result follows. Thus we focus on the case for $t \geq 1$. As in the lecture notes, we also only focus on showing the bound for I_1 . The bound for I_3 will follow nearly identically. As hinted in the problem, define

$$\varphi = \frac{2\pi x\xi}{t} + \sqrt{4\pi^2\xi^2 + 1},$$

and assume that $\partial\varphi/\partial\xi = 0$. I.e.,

$$0 = \frac{\partial\varphi}{\partial\xi} = \frac{2\pi x}{t} + \frac{1}{2}(4\pi^2\xi^2 + 1)^{-1/2} \cdot 8\pi^2\xi \implies \frac{\pi^2 x^2}{t^2} = \frac{4\pi^4\xi^2}{4\pi^2\xi^2 + 1} \implies \xi^2 = \frac{x^2}{4\pi^2(t^2 - x^2)}.$$

Moreover, we compute,

$$\begin{aligned} \partial_\xi^2 \varphi &= \frac{4\pi^2}{\sqrt{4\pi^2\xi^2 + 1}} - \frac{4\pi^2\xi}{(4\pi^2\xi^2 + 1)^{3/2}} \cdot 8\pi^2\xi = \frac{4\pi^2}{\sqrt{4\pi^2\xi^2 + 1}} - \frac{16\pi^4\xi^2}{(4\pi^2\xi^2 + 1)^{3/2}} \\ &= \frac{4\pi^2}{(4\pi^2\xi^2 + 1)^{3/2}}. \end{aligned}$$

Now we consider the integral

$$I_1 = \int_{\mathbb{R}} \exp\left(it\left(\frac{2\pi x \cdot \xi}{t} + \sqrt{4\pi^2\xi^2 + 1}\right)\right) \hat{\phi}_0(\xi) d\xi = \int_{\mathbb{R}} e^{i\varphi t} \hat{\phi}_0(\xi) d\xi.$$

Taylor expanding φ around ξ_0 (where $\xi_0^2 = x^2/(4\pi(t^2 - x^2))$), we have

$$\varphi = \left(\frac{2\pi x\xi_0}{t} + \sqrt{4\pi^2\xi_0^2 + 1}\right) + \frac{2\pi^2}{(4\pi^2\xi_0^2 + 1)^{3/2}}(\xi - \xi_0)^2 + \dots$$

Therefore,

$$I_1 \approx \hat{\phi}_0(\xi_0) \exp\left(it \cdot \left[\frac{2\pi x\xi_0}{t} + \sqrt{4\pi^2\xi_0^2 + 1}\right]\right) \cdot \int_{\mathbb{R}} \exp\left(it \cdot \frac{2\pi^2}{(4\pi^2\xi_0^2 + 1)^{3/2}}(\xi - \xi_0)^2\right) d\xi$$

The latter integral, which we denote by J , evaluates to

$$J = \int_{\mathbb{R}} \exp\left(i \cdot \frac{2\pi^2(t^2 - x^2)^{3/2}}{t^2}(\xi - \xi_0)^2\right) d\xi = \left(\frac{2\pi}{|c|}\right)^{1/2} e^{i(\pi/4)},$$

where

$$c = \frac{4\pi^2(t^2 - x^2)^{3/2}}{t^2}$$

Therefore, one has that

$$|I_1| \approx |\hat{\phi}_0(\xi_0)| \cdot \left(\frac{t^2}{2\pi(t^2 - x^2)^{3/2}}\right)^{1/2} \sim \frac{1}{t^{1/2}}$$

where we use the fact that $\hat{\phi}_0$ is Schwartz, and thus, bounded. For $t > 1$, we have $1 + t \leq 2t \implies (1+t)^{1/2} \leq \sqrt{2}\sqrt{t} \implies \frac{\sqrt{2}}{(1+t)^{1/2}} \geq \frac{1}{t^{1/2}}$. Thus, $|I_1| \lesssim \frac{1}{(1+t)^{1/2}}$, where the constant was dependent only on the initial data. Note that we used the idea of *stationary phase approximations*¹.

¹https://en.wikipedia.org/wiki/Stationary_phase_approximation

(c) We can simply generalize the entire above argument to show that the desired decay rate is satisfied.

Problem E.1.9. Show that the bound $\sup_{t,x} |\partial\phi| \leq C$ for the solutions to the wave equation in $\mathbb{R} \times \mathbb{R}$ with smooth and compactly supported data is sharp in the following sense: suppose the initial data are given by $(\phi, \partial_t\phi)|_{\{t=0\}} = (\phi_0, \phi_1) \in C_c^\infty \times C_c^\infty$. Then if there exists a non-increasing function $A(t)$ such that $\lim_{t \rightarrow \infty} A(t) = 0$ and

$$(\sup_x |\partial\phi|)(t) \leq A(t),$$

then $\phi_0 = \phi_1 = 0$.

Let ϕ be a solution to the linear wave equation in $\mathbb{R} \times \mathbb{R}$ with compactly supported smooth data, and such that $\sup_{t,x} |\partial\phi| \leq C$. We recall d'Alembert's formula, which states that

$$\phi(t, x) = \frac{1}{2} [\phi_0(x+t) + \phi_0(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} \phi_1(y) dy.$$

We then compute,

$$|\partial\phi|^2(t, x) = \frac{1}{2} [(\phi'_0)^2(x+t) + (\phi'_0)^2(x-t) + \phi_1^2(x+t) + \phi_1^2(x-t)] + \phi'_0(x+t)\phi_1(x+t) - \phi'_0(x-t)\phi_1(x-t).$$

Let $[a, b]^{\text{cpt}} \subseteq \mathbb{R}$ be a compact interval large enough to contain the support of the initial data. Fix an arbitrary point $x_0 \in [a, b]$, and let $x_1(t) = x_0 + t$ and $x_2(t) = x_0 - t$. Then it is easily shown that

$$\lim_{t \rightarrow \infty} |\partial\phi|^2(t, x_1(t)) = \frac{1}{2} [(\phi'_0)^2(x_0) + (\phi_1)^2(x_0) - \phi'_0(x_0)\phi_1(x_0)]$$

and

$$\lim_{t \rightarrow \infty} |\partial\phi|^2(t, x_2(t)) = \frac{1}{2} [(\phi'_0)^2(x_0) + (\phi_1)^2(x_0) + \phi'_0(x_0)\phi_1(x_0)].$$

By our hypothesis on the decay rate, we must also have $\lim_{t \rightarrow \infty} |\partial\phi|^2(t, x) = 0$ for every fixed $x \in \mathbb{R}$. Thus both of the above limits are zero. This yields $\phi'_0(x_0), \phi_1(x_0) = 0$. But since $x_0 \in [a, b]$ was arbitrary, $\phi'_0, \phi_1 \equiv 0$. Thus $\phi_0(x) \equiv c$ is a constant. But since the energy decays to zero as $t \rightarrow \infty$, but energy is also conserved, we must have $c \equiv 0$. Thus the conclusion follows.

Problem E.1.12. Show using the fundamental solution that solutions to the linear wave equation in $(1+1)$ -dimensions; i.e., $\mathbb{R} \times \mathbb{R}$, satisfy

$$\sup_{t,x} |\partial\phi| \leq C \sup_x |\partial\phi|(0, x)$$

for some $C > 0$ independent of ϕ . Show that the corresponding estimate fails in $\mathbb{R} \times \mathbb{R}^3$ (in fact, all dimensions $n \geq 2$!); i.e.,

$$\sup_{\{(\phi_0, \phi_1) : \sup_x |\partial\phi|(0, x) = 1\}} \left(\sup_{t,x} |\partial\phi| \right) = \infty.$$

This shows that unlike the L^2 norm for the derivatives of ϕ , which is propagated by the equation in all dimensions, the L^∞ norm for the derivative of ϕ is not propagated.

2.4 Random Computations

Problem 2. Analyze the modified energy, $E_p(t)$, defined by: for each $p \in (0, \infty)$, define

$$E_p(t) := \|\partial_t \phi\|_{L^p(\mathbb{R}^n)}^p(t) + \|\nabla \phi\|_{L^p(\mathbb{R}^n)}^p(t).$$

For $p = \infty$, define

$$E_\infty(t) = \|\partial_t \phi\|_{L^\infty(\mathbb{R}^n)}(t) + \|\nabla \phi\|_{L^\infty(\mathbb{R}^n)}(t).$$

Let ϕ be a sufficiently regular solution to the linear, homogeneous wave equation $\square \phi = 0$ on $\mathbb{R}^{1+n}$, and for $0 < p < \infty$, let

$$E_p(t) = \int_{\mathbb{R}^n} |\partial_t \phi|^p + |\nabla \phi|^p dx,$$

and for $p = \infty$, let

$$E_\infty(t) = \|\partial_t \phi\|_{L^\infty(\mathbb{R}^n)} + \|\nabla \phi\|_{L^\infty(\mathbb{R}^n)}.$$

We have already seen that when $p = 2$, $E_p(t)$ is constant in time for all $n \geq 1$. Our goal is to see if this conservation law extends to other p . First, let us consider $p = \infty$. We recall the pointwise dispersive estimates,

$$\sup_{x \in \mathbb{R}^n} |\phi(x, t)| \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}}.$$

Since derivatives commute, we have that $\partial_t \phi$ and $\nabla \phi$ are both solutions to the linear, homogeneous wave equation. So in fact, one has

$$\sup_{x \in \mathbb{R}^n} |\partial_t \phi(x, t)|, \sup_{x \in \mathbb{R}^n} |\nabla \phi(x, t)| \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}}.$$

Therefore for $n \geq 2$,

$$E_\infty(t) = \|\partial_t \phi\|_{L^\infty(\mathbb{R}^n)} + \|\nabla \phi\|_{L^\infty(\mathbb{R}^n)} \lesssim \frac{1}{(1+t)^{\frac{n-1}{2}}} \rightarrow 0 \text{ as } t \rightarrow \infty,$$

so that $E_\infty(t)$ is not constant in time. Now consider $n = 1$. We can easily disprove the conservation law via an example. Consider the Cauchy wave problem,

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = ((-1+x^2)\chi_{[-1,1]}, 0) \end{cases}$$

Then $E_\infty(0) = \|\nabla \phi\|_{L^\infty}(0) = 2$. However, in the period $0 < t < 1/2$, $\|\nabla \phi\|_{L^\infty(\mathbb{R})} > 1$ so that $E_\infty(t) > 2$ for all $t \in (0, 1/2)$. Thus $E_\infty(t)$ cannot be conserved. This rules out any energy conservation law for $p = \infty$.

Now let us consider $p = 1$,

$$E_1(t) = \int_{\mathbb{R}^n} |\partial_t \phi| + |\nabla \phi| dx = \|\partial_t \phi\|_{L^1(\mathbb{R})} + \|\nabla \phi\|_{L^1(\mathbb{R})}.$$

We will consider the case $n = 1$ first. Consider the Cauchy wave problem,

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = ((-1+|x|)\chi_{[-1,1]}, 0) \end{cases}$$

Then $E_1(0) = \|\nabla \phi\|_{L^1(\mathbb{R})} = 2$. Now let $t \in (0, 1/2)$. Then

$$\begin{aligned} \|\nabla \phi\|_{L^1(\mathbb{R})}(t) &= \int_{-1-t}^{-1+t} \frac{1}{2} dt + \int_{-1+t}^{-t} dt + \int_t^{1-t} dt + \int_{1-t}^{1+t} \frac{1}{2} dt \\ &= 2 - 2t. \end{aligned}$$

Likewise, we compute $\|\partial_t \phi\|_{L^1(\mathbb{R})}(t) = 4t$. Therefore, $E_1(t) = 4t + 2 - 2t = 2 + 2t > 2$. Evidently, $E_1(t)$ is not, in general, conserved in $n = 1$. Now we consider $n \geq 2$. We modify the Cauchy problem from above as follows,

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = ((-1 + |x|)\chi_{\{|x| \leq 1\}}, 0), \end{cases}$$

where $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. Then we compute $E_1(0) = \|\nabla \phi\|_{L^1(\mathbb{R}^n)}(0) = \omega_n$, where ω_n is the volume of the unit n -ball. On the other hand, for $t \in (0, 1/2)$, switching to radial coordinates and using the dispersion of solutions to the wave equation, we can show that the energy exceeds the initial energy. Thus $E_1(t)$ is not conserved for any $n \geq 2$. This rules out any energy conservation law for $p = 1$. Going one step further, we have the following.

Definition 2.1 (Energy). *Let $n \geq 1$ be a positive integer, and let ϕ be a solution to the linear, homogeneous wave equation*

$$\begin{cases} \square \phi = 0, \\ (\phi, \partial_t \phi) \upharpoonright_{\{t=0\}} = ((-1 + |x|)\chi_{\{|x| \leq 1\}}, 0) \end{cases} \quad (1)$$

in $\mathbb{R}^{1+n}$. For each $p \in (0, \infty)$, define

$$E_p(t) := \|\partial_t \phi\|_{L^p(\mathbb{R}^n)}^p(t) + \|\nabla \phi\|_{L^p(\mathbb{R}^n)}^p(t).$$

For $p = \infty$, define

$$E_\infty(t) = \|\partial_t \phi\|_{L^\infty(\mathbb{R}^n)}(t) + \|\nabla \phi\|_{L^\infty(\mathbb{R}^n)}(t).$$

Conjecture. *Let $n \geq 1$ be a positive integer, and let ϕ be a (nontrivial) solution to (1). For all $p \in (0, \infty] \setminus \{2\}$,*

$$\sup_{t \in (0, 1/2)} E_p(t) > E_p(0).$$

If $p = 2$, then $E_p(t)$ is constant for all $t \geq 0$.